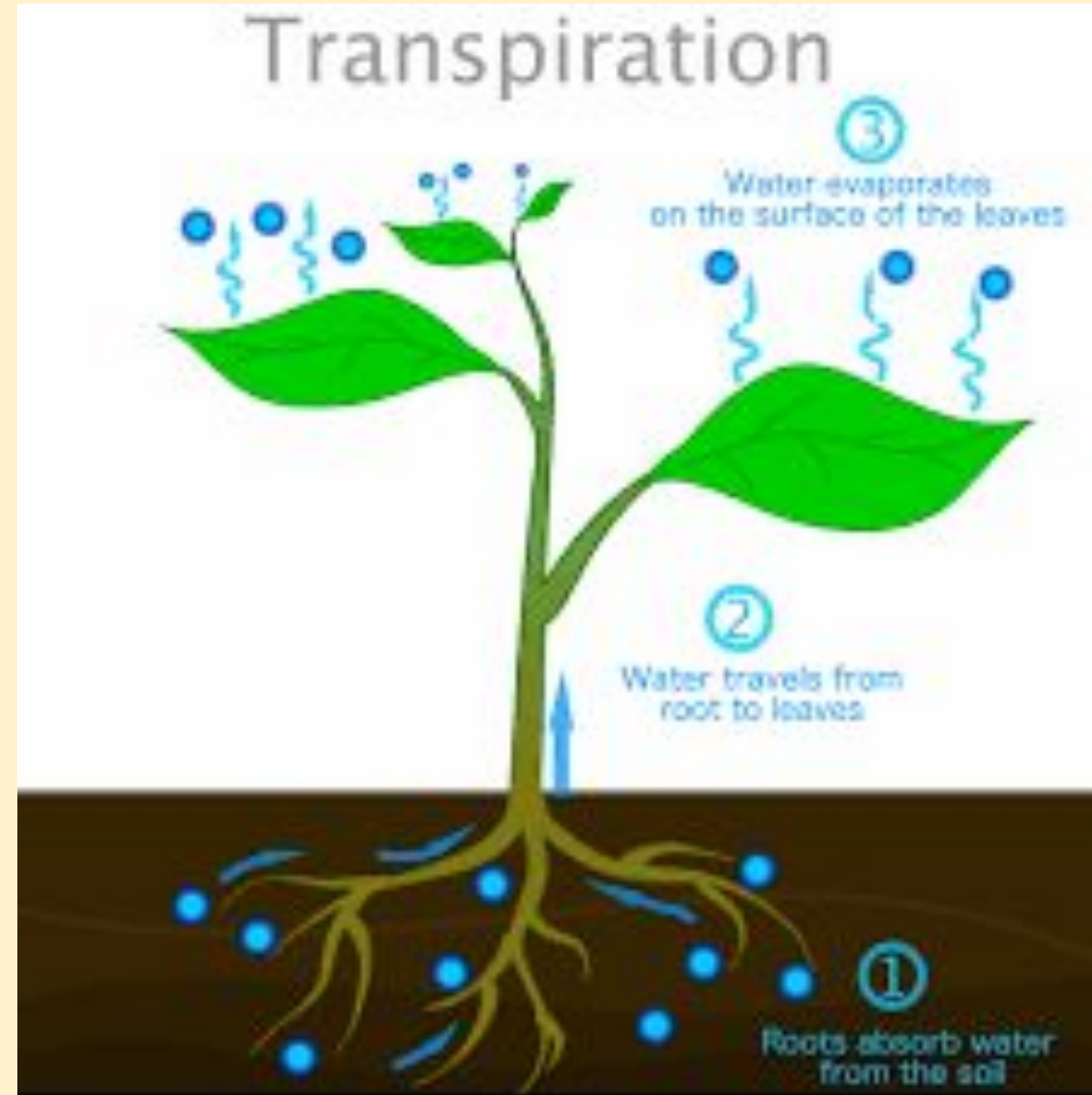


Starter HANGMAN



Starter HANGMAN

TRANSPIRATION



Task: Create a 'wanted' poster

Research one desert animal to create a wanted poster for.

Include:

- What it eats and what eats it
- The habitat it lives in
- How the species is adapted to survive in its habitat - Adaptations
- How its actions affect other organisms or non-living parts of the ecosystem (not just feeding)
- How other organisms or non-living things affect it

Wanted: Fenec Fox!



Name: Fenec Fox

Description of Animal: Fenec Foxes are the smallest of all foxes. They are 14-17 inches long when full grown. They weigh between 2 and 3.5 pounds. They have huge ears and large eyes. Their fur is sandy-colored.

Where Does it Live? The Sahara Desert, and other sandy deserts.

What Does it Eat? They are omnivores. They eat plants and animals. Some of the things they eat are: berries, fruit, small mammals, birds, eggs and lizards.

Adaptations:

1. Big Ears: Helps his body cool off.
2. Fur on the bottom of his feet: Helps keep his feet from burning on the desert sand.
3. Nocturnal: He comes out at night when it is cooler.
4. Color of Fur: Blends in with desert sand
5. Large eyes: Helps him see well at night.

Wanted Poster Feedback

Look at each others completed wanted posters

Give feedback to each of the posters. Your feedback should include:

WWW - what went well

What have they done well? What bit of information is most interesting?

EBI - even better if

Is something missing? What questions do you have that could have been included?

What is a native species?

Which of the pictures are native species to Malaysia?

What do we call the ones that aren't native?

Native species =
本土物种
토착종



What is a native species?

Which of the pictures are native species to Malaysia?

**What do we call the ones that aren't native?
How did they get there?**

We call them

Introduced

Invasive

**Or
Intruders**

What's the reason for each name? Discuss.



What is a native species
Which of the pictures are native species to Malaysia?

What do we call the ones that aren't native?
How did they get there?

We call them

Introduced. - by humans from one country or continent to another

Invasive – often the introduced species ‘invade’ – take over territory of another animals / eat the local wildlife / cause a lot of damage in their new ‘home’.

Or
Intruders

....so it's yet another environmental issue caused by us humans!



Title: Invasive Species

- A species that has been introduced or is not native to an environment
- Invasive species are generally harmful to the local habitat or native species
- Unwanted pest

Invasive species =
入侵物种
침입종



Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
Invasive Species	I can name an invasive species and describe what a native species is.	I can describe how invasive species affect native plants and animals in an ecosystem.	I can explain why controlling invasive species is important and evaluate different methods of control.
Case Study - New Zealand (Debate)	I can identify an invasive species problem in New Zealand.	I can describe the ecological impact of invasive species and outline arguments for control.	I can construct and defend an argument in a debate using evidence about species control and conservation.
Bioaccumulation	I can state what an insecticide is and give a simple example.	I can describe how chemicals like DDT build up in organisms and impact the food chain.	I can explain why organisms at the top of the food chain have higher toxin levels and the environmental consequences.
Weather and Climate	I can describe the difference between weather and climate.	I can explain how weather affects plants and animals in an ecosystem.	I can analyse weather data to predict effects on local ecosystems and seasonal changes.
Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.

Intruders in an Ecosystem

What is a native species?

Which of the pictures are native species to Malaysia?



Examples of Invasive Species

- **Grey squirrel**

Introduced to the UK from America, has displaced the native **red squirrel** by competing for food resources. It is also a carrier of disease that kills red squirrels

Invasive species =
入侵物种
침입종



Examples of Invasive Species

- **American beaver**

Introduced to eastern Russia and Argentina. Introduced beavers can cause problems by damming up rivers and felling trees, damaging forest

Invasive species =
入侵物种
침입종



Examples of Invasive Species

•Cane toad

Deliberately introduced to Australia to control another pest (a beetle that fed on sugar cane) but it didn't and has spread dramatically, poisoning animals that predate on them and displace native toads

Invasive species =
入侵物种
침입종



How do they invade?

Most invasions are due to human activities:

- Plant or pet trade
 - Commercial shipping
 - Tourism
- ...and many other routes...

What makes a good invader?

Talk to the person next to you, what traits do you think most invasive species would have?

- Rapid reproduction
- Fast growth
- Tolerant to many different conditions
- Able to live off many different foods
- Used by humans

Invader =
侵略者
침략자

3:00

But why is it a problem?

Impacts of invasive species

- Reduce native **biodiversity**
- Predation on local species
- Spread diseases
- Upset balance of local ecosystem
- Agricultural pests are very expensive!!

Biodiversity =
生物多样性
생물 다양성

Case Study: New Zealand

- NZ separated from all other land 66 million years ago
- Before humans arrived in NZ, there were **no mammal predators**
- This meant the birds didn't have to evolve any adaptations to fight off these predators!
- The result: many defenseless ground nesting birds that you can only find in NZ!



Case Study: New Zealand

The result: many defenseless ground nesting birds that you can only find in NZ!



Kiwi



Kokako



This Huia is now extinct



Kakapo

Case Study: New Zealand

These days there are many mammals in NZ, some of these are harmful to the bird populations!

- Stoats
- Rats
- Possums

Scientists think that 53% of native birds species have already gone extinct in NZ since humans arrived

How can we control invasive species?

- Culling
- Trapping
- Removing plants
- Poison
- Insecticides
- Using other living organism that eat the pest



Case Study: New Zealand

- There are now BIG conservation efforts in NZ to try to protect the remaining native bird species.
- This means trying to eradicate (get rid of) stoats and other mammal predators

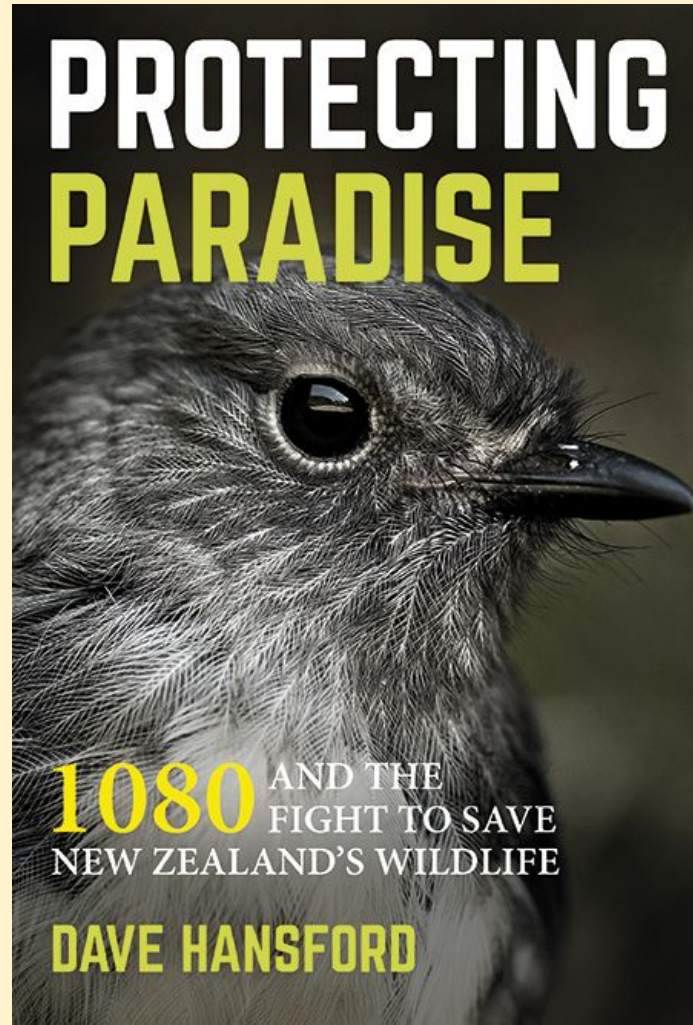
How?

Many traps are laid out in the forests to trap mammals

Large areas are covered with poison (called 1080) that kills mammal species

Case Study: New Zealand

BUT - not everyone agrees with this! - what do you think?



australian
animal
poisons
centre

Why do some introduced species cause problems?

Stoats normally live in Europe. Stoats are not a 'problem animal' in the ecosystems where they normally live.

- With your partner, think of ideas to explain why stoats are such a problem in NZ, but not a problem in their native countries in Europe

Be ready to share your ideas with the class!



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Why do some introduced species cause problems?

Summary:

Stoats aren't a problem in Europe because they are part of a balanced ecosystem with predators, competitors, and prey that can escape them.

In New Zealand, the ecosystem evolved without mammals, so stoats suddenly became a top predator with nothing to stop them — making them extremely dangerous to native wildlife.



Research Task

In groups of 3 or 4 research another example of an invasive species (it can be in New Zealand or any other country!)

Create a short role play to explain what happened

